

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415555
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Armbruster; Patrick et al.

Cleaning cart

Abstract

The invention relates to cleaning cart, comprising a frame having a frame lower part, a frame upper part and support parts which connect the frame lower part to the frame upper part and are spaced apart from one another, rollers at the frame lower part for moving on a ground surface, storage devices for receiving therein and/or supporting thereon cleaning accessories, retaining elements for holding the storage devices, and receiving elements arranged at the support parts in a predefined position, wherein the retaining elements are selectively fixable to the receiving elements for having the storage devices assume different relative positions with respect to the support parts.

Inventors: Armbruster; Patrick (Winnenden, DE), Burchard; Jochen (Winnenden, DE), Dammköhler; Denis (Winnenden, DE), Dobler; Kamila (Winnenden, DE), Guegercin; Beyza (Winnenden, DE), Jahn; Marten (Winnenden, DE), Kuhn; Jens (Winnenden, DE), Maniscalco; Calogero (Winnenden, DE)

Applicant: ALFRED KÄRCHER SE & CO. KG (Winnenden, DE)

Family ID: 1000008819798

Assignee: ALFRED KÄRCHER SE & CO. KG (Winnenden, DE)

Appl. No.: 18/050369

Filed: October 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230182796 A1	Jun. 15, 2023

Foreign Application Priority Data

DE	10 2020 111 731.0	Apr. 29, 2020
----	-------------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/EP2021/058284 20210330 PENDING child-doc US 18050369

Publication Classification

Int. Cl.: B62B3/00 (20060101); A47L13/51 (20060101); B62B5/00 (20060101)

U.S. Cl.:

CPC B62B3/005 (20130101); A47L13/51 (20130101); B62B3/004 (20130101); B62B5/0096 (20130101); B62B2202/50 (20130101)

Field of Classification Search

CPC: B62B (3/005); B62B (3/004); B62B (5/0096); B62B (2202/50); A47L (13/51)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10029718	12/2017	Benning	N/A	B62B 3/003
2011/0133417	12/2010	Rouillard et al.	N/A	N/A
2014/0049014	12/2013	Schumacher et al.	N/A	N/A
2016/0332651	12/2015	Benning	N/A	B62B 3/003
2018/0103820	12/2017	Menzel et al.	N/A	N/A
2023/0182796	12/2022	Armbruster	280/47.35	A47L 13/51

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202445473	12/2011	CN	N/A
206166275	12/2016	CN	N/A
210300213	12/2019	CN	N/A
3247135	12/1983	DE	N/A
8430997	12/1984	DE	N/A
10047352	12/2001	DE	N/A
20-2004-000404	12/2003	DE	N/A
1190664	12/2001	EP	N/A
1437283	12/2003	EP	N/A
1512606	12/2004	EP	N/A
2314497	12/2010	EP	N/A
3072776	12/2015	EP	N/A
3208174	12/2016	EP	N/A
2716661	12/1994	FR	N/A

MI20100965	12/2010	IT	N/A
WO-2010051841	12/2009	WO	A47B 81/02
WO-2021165520	12/2020	WO	A47L 9/0009
WO-2021219313	12/2020	WO	A47L 13/51

OTHER PUBLICATIONS

International Search Report and Written Opinion for International (PCT) Patent Application No. PCT/EP2021/058284, dated Nov. 4, 2021, 13 pages. cited by applicant

Search Report for Germany Patent Application No. 10-2020-111731.0, dated Apr. 29, 2020, 7 pages. cited by applicant

Primary Examiner: Meyer; Jacob B

Attorney, Agent or Firm: Sheridan Ross P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of international application number PCT/EP2021/058284, filed on Mar. 30, 2021, and claims the benefit of German application number 10 2020 111 731.0, filed on Apr. 29, 2020, which are incorporated herein by reference in their entirety and for all purposes.

FIELD OF THE INVENTION

(1) The present invention relates to a cleaning cart, comprising a frame having a frame lower part, a frame upper part and support parts which connect the frame lower part to the frame upper part and are spaced apart from one another, rollers at the frame lower part for moving on a ground surface, and storage devices for receiving therein and/or supporting thereon cleaning accessories.

BACKGROUND OF THE INVENTION

(2) Despite modern cleaning machines, in particular in the field of interior cleaning, some of which can be automated, cleaning carts are of great importance in the field of daily cleaning. A particular emphasis is on cleaning tasks that are to be manually performed in larger buildings, such as office buildings, hotels, nursing facilities, hospitals, airports, event buildings, or the like. The cleaning carts serve to carry cleaning utensils. Examples of these include, in particular, manually guided cleaning machines and cleaning tools, receptacles for storing cleaning accessories, cleaning liquids or cleaning chemicals, receptacles for receiving dirty liquid or waste, cleaning cloths, or other items for surface cleaning.

(3) The cleaning carts comprise rollers so that they can be moved, usually hand-guided, by a user over a ground surface. In the present case, the rollers can be stationary, i.e., unsteered.

Alternatively, provision can be made for the rollers to be steered rollers. A combination of unsteered rollers and steered rollers can also be provided.

(4) The cleaning carts comprise a frame having a frame lower part and a frame upper part spaced apart therefrom, these being connected to one another via support parts, wherein the support parts are, for example, profile parts. For example, the frame can define a receiving space for the cleaning utensils, which receiving space can be closed, for example, by way of side walls, a top wall and/or doors. However, this is not an absolute requirement. Correspondingly, a corpus or a rack in particular can also be considered as a frame in the present case.

(5) An object underlying the present invention is to provide a cleaning cart that has increased versatility.

SUMMARY OF THE INVENTION

(6) In an aspect of the invention, a cleaning cart is provided. The cleaning cart comprises a frame having a frame lower part, a frame upper part and support parts which connect the frame lower part to the frame upper part and are spaced apart from one another, rollers at the frame lower part for moving on a ground surface, storage devices for receiving therein and/or supporting thereon cleaning accessories, retaining elements for holding the storage devices, and receiving elements arranged at the support parts in a predefined position. The retaining elements can be selectively fixed to the receiving elements for having the storage devices assume different relative positions with respect to the support parts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing summary and the following description may be better understood in conjunction with the drawing figures, of which:
- (2) FIG. 1 illustrates a perspective view of the cleaning cart in accordance with the invention in a preferred embodiment;
- (3) FIG. 2 illustrates the cleaning cart of FIG. 1 after doors, side walls and a top wall have been removed;
- (4) FIG. 3 illustrates the cleaning cart of FIG. 2 after containers have been removed;
- (5) FIG. 4 illustrates a side view of the cleaning cart, looking in the direction of arrow “4” of FIG. 3;
- (6) FIG. 5 illustrates a sectional view along line 5-5 of FIG. 4;
- (7) FIG. 6 illustrates a perspective partial view of detail A of FIG. 5, showing a support part and a receiving part of the cleaning cart in exploded representation;
- (8) FIG. 7 illustrates a perspective representation of the receiving part;
- (9) FIG. 8 illustrates an enlarged representation of detail B of FIG. 7;
- (10) FIG. 9 illustrates a perspective representation of a retaining element of the cleaning cart of FIG. 1;
- (11) FIG. 10 illustrates an enlarged detail representation, looking in the direction of arrow “10” of FIG. 9;
- (12) FIG. 11 illustrates a sectional view along line 11-11 of FIG. 4;
- (13) FIG. 12 illustrates an enlarged representation of detail A of FIG. 5;
- (14) FIG. 13 illustrates a representation corresponding to FIG. 12, wherein a retaining element assumes a different orientation;
- (15) FIG. 14 illustrates a representation corresponding to FIG. 2, wherein the containers assume extended positions with respect to a frame of the cleaning cart;
- (16) FIG. 15 illustrates a representation similar to FIG. 14, wherein the containers assume different extended positions and retaining elements for a container have a different orientation;
- (17) FIG. 16 illustrates a representation similar to FIG. 15, wherein two containers have been replaced by a different container and retaining elements have been removed;
- (18) FIG. 17 illustrates, in top view, a frame lower part of the cleaning cart;
- (19) FIG. 18 illustrates, in bottom view, a frame upper part of the cleaning cart;
- (20) FIG. 19 illustrates another perspective representation of the cleaning cart of FIG. 1 with a door open, shown in a partially exploded representation;
- (21) FIG. 20 illustrates an enlarged representation of detail C of FIG. 19 in a simplified, exploded view;
- (22) FIG. 21 illustrates an enlarged representation of detail D of FIG. 19 in a simplified, exploded view;
- (23) FIG. 22 illustrates a representation similar to FIG. 19, wherein the door is hinged to the

cleaning cart on a different side and is shown in an open position;

(24) FIG. 23 illustrates another representation of the cleaning cart of FIG. 1, wherein a segment of a side wall has been removed and another segment of the side wall is shown as having been displaced;

(25) FIG. 24 illustrates an enlarged representation of detail E of FIG. 23;

(26) FIG. 25 illustrates an enlarged representation of detail F of FIG. 23;

(27) FIG. 26 illustrates, in perspective representation, a further preferred embodiment of the cleaning cart in accordance with the invention;

(28) FIG. 27 illustrates an enlarged view of detail G of FIG. 26 after removal of a door wall and in a partially exploded representation;

(29) FIG. 28 illustrates a perspective partial representation of the cleaning cart, looking in the direction of arrow "18" of FIG. 26;

(30) FIGS. 29 and 30 illustrate perspective representations of a further preferred embodiment of the cleaning cart in accordance with the invention; and

(31) FIG. 31 illustrates a perspective representation of a further preferred embodiment of the cleaning cart in accordance with the invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

(32) Although the invention is illustrated and described herein with reference to specific embodiments, the invention is not intended to be limited to the details shown. Rather, various modifications may be made in the details within the scope and range of equivalents of the claims and without departing from the invention.

(33) The present invention relates to a cleaning cart, comprising a frame having a frame lower part, a frame upper part and support parts which connect the frame lower part to the frame upper part and are spaced apart from one another, rollers at the frame lower part for moving on a ground surface, storage devices for receiving therein and/or supporting thereon cleaning accessories, retaining elements for holding the storage devices, and receiving elements arranged at the support parts in a predefined position, wherein the retaining elements are selectively fixable to the receiving elements for having the storage devices assume different relative positions with respect to the support parts.

(34) In the cleaning cart in accordance with the invention, storage devices can be fixed to the support parts via the retaining elements and the receiving elements. In particular, here, the possibility exists for the storage devices to be fixed in different relative positions with respect to the support parts. In the present case, this can be realized by virtue of the fact that retaining elements can cooperate with different receiving elements in order to achieve different relative positions. This gives the possibility of providing the cleaning cart in different configurations depending, for example, on the cleaning task and/or preferences of a user, wherein the storage devices are brought into a preferred position at the frame. It is conceivable for the positioning of the storage devices to be done at the factory. However, it is advantageous in particular if the user himself or herself can arrange the storage devices in the desired positions. To this end, the retaining elements are preferably releasable from the receiving elements.

(35) In particular, the present invention gives the possibility of a user-defined "modular-like" configuration and such an assembling by the user. For example, a quantity of identical and/or different storage devices can be provided, which can be selectively fixed to the frame as desired by the user. It is in particular conceivable to have more storage devices than are usually carried along with the cleaning cart or than can be attached thereto. In this way, for example, an ideal assembly of storage devices can be assembled, tailored to the particular cleaning task.

(36) Preferably, the retaining elements are freely selectively fixable to the receiving elements.

(37) It is advantageous for a plurality of receiving elements to be provided in spaced relation to one another in a height direction of the cleaning cart and for the retaining elements to be selectively fixable at different positions in the height direction. The height of the storage devices at the frame

can thereby be adjusted. By way of example, the retaining elements are attached between the frame lower part and the frame upper part.

(38) In a preferred embodiment of the invention, the receiving elements are arranged equidistantly from one another in the height direction. For example, a grid of receiving elements can be provided relative to the height direction.

(39) As used herein, positional and orientational terms, such as “bottom”, “top”, “height direction”, or the like, are to be understood to relate to an intended use of the cleaning cart. When in intended use, the cleaning cart is supported via its rollers on the ground surface. It can be assumed, without limitation, that the ground surface is oriented horizontally.

(40) It is advantageous if, at the support parts, receiving elements are provided which are arranged on two sides, oriented transversely and in particular perpendicularly to one another, of the support parts, and if the retaining elements can be selectively fixed to a receiving element on a first side or to a receiving element on a second side of the two sides, with respect to different orientations of the retaining elements relative to the support part. Depending on the receiving element of what side (first or second side) the retaining element is fixed to, the latter can be oriented differently relative to the support part.

(41) Advantageously, two orientations are provided, these being oriented transversely and in particular perpendicularly to one another. A respective orientation can, for example, define a pullout direction for an extendable storage device.

(42) For example, the first and second sides can be adjacent to one another and extend in the height direction, preferably wherein a plurality of receiving elements are provided in spaced apart relation to one another in the height direction.

(43) Preferably, the receiving elements arranged on the two sides arranged transversely to one another are arranged at the same position in a height direction of the cleaning cart.

(44) Preferably, the storage devices are or comprise containers which, for example, have or form a receiving space for cleaning utensils. By way of example, the containers are bucket-shaped or basin-shaped. Extendable containers can be drawer-like or form a drawer.

(45) For example, provision can be made for at least two containers to have different sizes, in particular wherein the containers have differing heights along a height direction of the cleaning cart, and preferably are of identical cross-section in a direction transverse to the height direction. This allows containers of differing sizes to be carried along as desired or as needed for the cleaning task.

(46) Provision can be made for at least two containers to be of identical size, and to be of identical configuration in particular.

(47) In a preferred embodiment, the containers can be substantially rectangular in cross-section in a direction transverse to a height direction of the cleaning cart.

(48) It can be advantageous for the cleaning cart to comprise a marking device for marking the storage devices, and specifically the containers. Via the marking device, the user is provided with information and guidance, perceivable by intuition, as to what purposes the storage devices are to be used for. By way of example, different containers can be provided on the one hand for clean or unused cleaning utensils and on the other for dirty or used cleaning utensils.

(49) For example, the marking device can be implemented by coloring. Here, for example, colored marking elements can be attached or are attachable to the storage devices, wherein the color is associated with the contents and/or purpose of the container.

(50) The storage devices, in particular the containers, can for example comprise a handle element graspable by the user. Preferably, the storage device can be carried and/or displaced relative to the frame by the handle element. By way of example, the handle element is a handle strip.

(51) It can be advantageous for the extensions of the containers in the height direction to have ratios to each other that are given by powers of two. For example, containers with a height ratio of two or with a height ratio of four can be provided.

(52) Alternatively or in addition, provision can be made for the extensions of the containers in the height direction to be integer multiples of each other.

(53) It is advantageous for the storage devices to comprise plate-shaped or substantially plate-shaped supporting elements having a respective supporting surface for cleaning utensils. A cleaning utensil can be placed on the supporting element. It is conceivable that, for example, a bucket-shaped receptacle be placed on a supporting element.

(54) In a preferred embodiment, the supporting elements can be intermediate shelves or dividing shelves of a receiving space of the cleaning cart.

(55) For example, the supporting elements can comprise, at a top side thereof, a depression enclosed by a rim to enhance the stability of cleaning utensils standing thereon.

(56) In particular, it can be advantageous for at least one supporting element to form a cover element for a container. In this way, the supporting elements can serve a twofold function.

(57) For example, provision can be made for cross-sections of the supporting element and of the container in a direction transverse to a height direction of the cleaning cart to be identical or substantially identical in order to completely or substantially completely cover the container.

(58) It is advantageous for a supporting element and a container underlying same in a height direction to be holdable to at least one retaining element, preferably wherein the supporting element directly covers the container. Separate retaining elements for the container and for the supporting element associated therewith as a cover for the container are thereby obviated. For example, the retaining element comprises guide members or retaining members arranged one over another for the container and the supporting element, these being configured as a groove or strip.

(59) In a preferred embodiment, the retaining elements are or comprise pullout guides, each of which has at least one guide member for the storage devices, in particular the containers and/or the supporting elements, wherein a respective storage device is displaceable along the guide member from a retracted position to at least one extended position and vice versa. For example, a container and/or a supporting element can be displaced from the retracted position in the receiving space of the cleaning cart to the at least one extended position in order to facilitate access to cleaning utensils for the user. In particular, the retracting and extending can also serve the mounting or dismounting of the storage devices.

(60) Provision can be made for not all of the storage devices of the cleaning cart to be extendable as described above. For example, only some of the storage devices are extendable.

(61) It is advantageous for the storage device, in particular the container and/or the supporting element, to be configured for assuming two extended positions and to be transferrable along the guide member starting from the retracted position in two mutually opposite directions to a respective extended position. For example, this allows access to the corresponding storage device from sides of the cleaning cart that face away from each other. To this end, the storage device can be pulled out in the respective desired direction.

(62) The storage devices, in particular the containers and/or the supporting elements, can, for example, comprise retaining projections which engage in the guide members, preferably wherein the respective storage device can be latched with the guide member when in the retracted position and/or in the at least one extended position. Via the latching, the storage devices can assume a defined position at the frame, which facilitates the handling of the cleaning cart for the user.

(63) For example, the guide members are groove-shaped and the retaining projections are strip-shaped.

(64) In an embodiment of simple construction, the pullout guides can be configured as moulded plastic parts for example.

(65) In a preferred embodiment, the pullout guides can be telescopic.

(66) It is advantageous for the storage devices, in particular the containers and/or the supporting elements, to be extendable relative to the frame along two directions oriented transversely and in particular perpendicularly to one another, depending on the orientation of the retaining elements

relative to the frame. For example, the pullout direction can be different depending on the cleaning task and preference of the user. For example, the two pullout directions are arranged in a common plane parallel to a plane of the frame lower part. By way of example, the storage devices can be pulled out, relative to a direction of travel of the cleaning cart, forwards and/or backwards (away from the user and in a direction towards the user respectively) and, where the retaining elements are differently oriented, to the left and/or to right (relative to the direction of displacement).

(67) Advantageously, the possibility exists for the user to orient the retaining elements relative to the frame along the first of the two orientations or alternatively along the second orientation and to fix them to the support parts. This allows the user to selectively implement storage devices that can be pulled out along the first orientation or along the second orientation.

(68) Preferably, the receiving elements are arranged on respective sides of the support parts that face towards another support part, spaced apart from the support part, in particular wherein at two support parts arranged a distance apart from each other, receiving elements are arranged at the support parts in opposing relation to one another. Preferably, here, the possibility exists in particular of positioning the retaining elements between the support parts, wherein at least one retaining element is arranged at each support part.

(69) In a preferred embodiment of the invention, a respective retaining element can be held at end sections thereof facing away from each other to two support parts spaced apart from each other. Here, a respective end section is held to one of the support parts.

(70) Preferably, the retaining elements in each case can be arranged in pairs, wherein a retaining element arranged at the frame in a height direction of the cleaning cart is associated with a corresponding retaining element in opposing relation thereto at the frame. For example, an extendable storage device, such as a container or a supporting element, can be held to the two retaining elements.

(71) Preferably, the frame forms a square shape in a top plan view. By way of example, the support parts are arranged at the four corners of the imaginary square.

(72) In accordance with the invention, the retaining elements can be fixed to the receiving elements which are arranged at the support parts. In particular, here, it is possible for the retaining elements to be fixed to the support parts directly or indirectly.

(73) In a preferred embodiment of the invention, the support parts comprise or form the receiving elements, in particular wherein the receiving elements are formed in one piece with the support parts. For example, this gives the possibility of fixing the retaining elements directly to the support parts.

(74) In a different preferred embodiment of the invention, it is advantageous for the cleaning cart to comprise receiving parts comprising the receiving elements, which receiving parts are formed separately from the support parts and are connected to the respective support parts. For example, this gives the possibility of fixing the retaining elements indirectly, via the receiving parts, to the support parts.

(75) Preferably, a plurality of receiving elements are arranged at the respective receiving part. Separate receiving parts can thereby be dispensed with, which facilitates the manufacture of the cleaning cart.

(76) Preferably, the support parts and the receiving parts can be optimally configured for their respective functions. For example, the support parts are configured as profile parts and/or are preferably made of a rugged material, for example metal.

(77) The receiving parts can, for example, be of strip-shaped configuration and can preferably be made of a plastics material.

(78) Preferably, the receiving parts can be fixed to the respective support part by a force-locking and/or form-locking connection.

(79) In particular, clamping and/or latching are conceivable for the fixing.

(80) Advantageously, the receiving parts can be fixed to the respective support part by hand and/or

without tools.

(81) In a preferred embodiment of the invention, the receiving parts can be slid into the respective support part. For example, the support part can have a groove-shaped recess or receiver into which the receiving part can be slid, wherein it is in engagement with the support part.

(82) Preferably, the receiving part can be slid into the support part in a longitudinal direction and can, when the cleaning cart is in its intended use, extend in the height direction thereof, for example. It is conceivable that a plurality of receiving parts be fixable to the support part and in particular slidable thereinto.

(83) It is advantageous for at least one retaining element to comprise two contact regions oriented transversely to each other and in particular perpendicularly adjacent to one another, via which contact regions the retaining element is in contact, preferably area contact, against corresponding contact regions of the support part or against corresponding contact regions of a receiving part comprising the receiving elements. In this way, a reliable relative orientation of the retaining element and the receiving elements can be ensured. For example, the support part or the receiving part and the retaining element are contacted against one another around corners, wherein the two sides are adjacent to one another.

(84) Preferably, the contact regions are planar in order to enable flange-like contact of the retaining element.

(85) Preferably, the retaining elements can be fixed to the receiving elements by a force-locking and/or form-locking connection.

(86) For example, provision can be made for the retaining elements to be clamped and/or latched with the receiving elements.

(87) It is advantageous for the retaining elements to be releasably fixable to the receiving elements so that the position of the retaining elements can be varied and/or unneeded retaining elements can be removed.

(88) The retaining elements can preferably be fixed to and/or released from the receiving elements by hand and/or without tools.

(89) In a preferred embodiment of the invention, provision can be made for the retaining elements and the receiving elements to comprise retaining projections and corresponding engagement openings for the retaining projections which are in engagement with one another when in a connecting position.

(90) For example, the retaining projections are arranged at the retaining elements, and the receiving elements comprise or form the engagement openings.

(91) Preferably, biasing elements are provided against whose action the retaining projections can be brought out of engagement with the engagement openings.

(92) For example, a biasing element is associated with a respective retaining projection in order to bias the retaining projection in a direction towards the connecting position. This ensures reliable engagement of the retaining projections in the engagement openings. Preferably, the retaining projection can be released from the engagement opening against the action of the biasing element.

(93) It is advantageous for the retaining elements and the receiving elements to comprise orienting projections and corresponding grooves for the orienting projections, these being in engagement with each other in the connected state of the retaining elements with the receiving elements. This enables reliable relative orientation of the retaining elements to the receiving elements.

(94) For example, the orienting projections are arranged at the retaining elements and the grooves are arranged at the support parts or at the receiving parts.

(95) Preferably, the orienting projections or grooves are formed separately from the retaining projections and the engagement openings.

(96) For example, receivers for the support parts can be formed at the frame lower part and/or at the frame upper part in order to facilitate assembly of the cleaning cart.

(97) As has already been mentioned, the cleaning cart can comprise a corpus which is formed by

the frame. For example, here, side walls and/or doors are provided. Preferably, the corpus encloses a receiving space in which the storage devices can be arranged.

(98) Correspondingly, the cleaning cart can comprise at least one door which can be transferred from a closed position to an open position and vice versa. For example, a storage space of the corpus can be accessed by opening the door.

(99) Preferably, the door comprises a door wall and fixing parts between which the door wall is held or arranged, wherein the fixing parts are pivotably held, directly or indirectly, to the frame via bearing elements. Advantageously, the bearing elements of the fixing parts define a pivot axis for the door. Preferably, the pivot axis extends in a height direction of the cleaning cart.

(100) A tension element can be provided for tensioning the fixing parts in place relative to one another.

(101) For example, in a preferred embodiment of the invention, a corresponding bearing element for a bearing element at the fixing part can be arranged or formed at the frame lower part and/or at the frame upper part.

(102) Provision can be made for a bearing element corresponding to a bearing element of a fixing part to be arranged or formed at at least one retaining element.

(103) The fastening mechanism for the door can be of identical or functionally equivalent configuration to the fastening mechanism for the retaining elements. For example, retaining projections are arranged at the door and corresponding engagement openings are arranged at the frame, these engaging with one another when in a connecting position. In this respect, reference may be had to what has been described in the foregoing, also with respect to the biasing elements.

(104) Preferably, the door can be attached to the cleaning cart in two different orientations and positions in which the door, rotating about parallel pivot axes in directions pointing away from each other relative to the frame, can be transferred to the open position. This lends high versatility to the cleaning cart. Here, provision can be made for a user to be allowed to alter the desired position of the door (the “hanging” of the door) on the cleaning cart.

(105) For example, provision can be made for the user, when facing the door, to swing the door open to the left (“left-hand opening door”) or to the right (“right-hand opening door”).

(106) To this end, for example, different bearing elements are provided at the frame lower part and/or at the frame upper part. Alternatively or in addition, at least one different retaining element having a respective bearing element is provided for this purpose in order to change the location of the door hinge.

(107) Preferably, the position of the door at the frame can be changed by hand and/or without tools.

(108) Preferably, arranged at the frame upper part is a top wall which can be transferred from a closed position to an open position and/or which can be removed from the frame upper part. This can create a reach-through opening through which access can be gained to the receiving space from the top. Alternatively or in addition, for example, a cleaning utensil can project upwardly from the receiving space and through the reach-through opening.

(109) Preferably, the cleaning cart is hand-guided and comprises a handle device graspable by a user.

(110) FIG. 1 shows an advantageous embodiment of the cleaning cart in accordance with the invention, designated by the reference numeral **10**. The cleaning cart **10** serves to carry cleaning utensils, in particular for manual cleaning, such as hand-guided cleaning machines or cleaning tools, containers for further cleaning utensils or cleaning accessories, cleaning liquids or cleaning chemicals, cleaning utensils such as wiping cloths, containers for holding waste, etc.

(111) The cleaning cart **10** can be moved on a ground surface **12** by a user in a hand-guided manner. Positional and orientational terms as used herein are assumed to relate to an intended use of the cleaning cart **10**. Here, the cleaning cart **10** is supported via rollers **14** on the ground surface **12** which, by way of example and not by way of limitation, can be considered to be horizontal.

(112) The cleaning cart **10** comprises a frame **16** which in the present case forms a corpus **18**. The

frame **16** comprises a substantially plate-shaped frame lower part **20** and a frame upper part **24** spaced relative thereto in a height direction **22**. The frame upper part **24** itself is frame-shaped and forms a reach-through opening **26**.

(113) The frame lower part **20** and the frame upper part **24** are connected to one another via support parts **28**, these being fixed at respective corner regions of the frame lower part **20** and the frame upper part **24**. To this end, receivers **30** can be provided at the lower frame part **20** and the upper frame part **24** (FIGS. **17** and **18**), into which receivers **30** the support parts **28** can be inserted. The support parts **28** are formed as profile parts.

(114) In the present case, four support parts **28** are provided, each spaced apart one from the other.

(115) The frame **16** forms a square shape in a top plan view. Here, the support parts **28** are arranged at the corners of the imaginary square.

(116) The rollers **14** are held to the lower frame part **20** and are, in the present case, configured as steerable rollers. Also conceivable is the use of unsteered rollers.

(117) Provided at the frame upper part **24** is a handle device **32** for the user to move the cleaning cart **10**. Usually, this can be realized in a forward movement along a travel direction **34**. The cleaning cart **10** has a front side **38** in the travel direction **34**. The handle device **32** is arranged on the rear side **40**.

(118) A transverse direction **36** is oriented transversely to the travel direction **34**, in particular wherein the travel direction **34**, the transverse direction **36** and the height direction **22** are pairwise perpendicular to one another.

(119) The cleaning cart **10** further comprises a left side **42** and a right side **44**.

(120) The frame **16** defines a receiving space **46** in which storage devices **48** for cleaning utensils are arranged. The receiving space **46** is closed by the corpus **18**. To this end, the cleaning cart **10** comprises two doors **50**, of which only one door **50**, on the left side **42**, is illustrated in the drawing.

(121) Furthermore, the cleaning cart comprises two side walls **52** (FIGS. **1**, **19**, **22** and **23**). The side walls **52** each have segments **54** arranged one above the other.

(122) On the top side, the cleaning cart **10** comprises a top wall **56** by which the reach-through opening **26** can be closed off. In particular, the top wall **56** can be releasably fixable to the frame upper part **24**. It is in particular desirable for the top wall **56** to be for example pivotable on the frame upper part **24** from a closed position to an open position and vice versa.

(123) The storage devices **48** are configured and adapted for receiving therein and/or supporting thereon cleaning accessories. The storage devices **48** in the present case comprise containers **58** and supporting elements **60**.

(124) The containers **58** themselves constitute cleaning accessories and are of a bucket-shaped or basin-shaped configuration. They have a receiving space for cleaning accessories. In the present case, the containers **58** have a substantially rectangular cross-section in the height direction **22**.

(125) In particular, the cleaning cart **10** can comprise a plurality of containers **58**. Of these, at least two containers **58** can be of identical configuration and/or at least two containers **58** can have different configurations.

(126) Provision can be made for the cleaning cart **10** to comprise more containers **58** than could be attached thereto and carried thereby. Depending on what particular containers **58** and, correspondingly, what particular supporting elements **60** are assembled, the cleaning cart **10** can be customized to cleaning tasks and/or preferences of the user.

(127) Preferably, different containers **58** only differ in their extensions along the height direction **22** and are otherwise of substantially identical configuration. In particular, for example, they can be substantially identical in their cross-section transverse to the height direction **22**. Differently from this, containers **58** formed to taller heights can, for example, have a greater taper because of their greater height.

(128) As shown in the examples of FIGS. **2** and **14**, the cleaning cart **10** can comprise two different

types of containers **58**, namely a tall container **58** and two less tall containers **58**.

(129) Provision can be made for the cleaning cart **10** to comprise further containers **58**. The example of FIG. **16** illustrates a further tall container **58**.

(130) It is advantageous for the containers **58**, and in a corresponding manner the supporting elements **60**, to be combined with each other in a modular-like manner, depending on the cleaning task and/or preference of the user.

(131) This is made possible by the ways and manners in which the storage devices **48** are held to the frame **16**.

(132) The containers **58** each include retaining projections **62** on opposite sides thereof. The retaining projections **62** are of strip-shaped configuration.

(133) Furthermore, in the present case, the containers **58** comprise handle elements **64**, likewise on opposite sides thereof, albeit where the retaining projections **62** are not arranged. The handle elements **64** are also strip-shaped.

(134) The cleaning cart **10** can comprise a marking device for marking the containers **58**. By way of example, the marking device comprises marking elements that can be attached or can be attachable to the containers **58**. For example, the marking element is arranged at the handle element **64**. The marking device can be implemented, for example, by coloring.

(135) The supporting elements **60** are plate-shaped, as can be seen from FIG. **3** for example. Preferably, here, on a top side of a respective supporting element **60**, a depression **68** is provided which is enclosed by a rim **66** and has a supporting surface for cleaning utensils.

(136) The support parts **28** are of longitudinally extending configuration and are formed from a metal material for example. On their outer side, the support parts **28** have an arcuate section **76** which substantially extends over a quarter circle. Projecting from the section **76** are two lateral sections **78**, on a respective side of the support part **28**.

(137) On a side facing away from the section **76**, the sections **78** transition into a connecting section **80** which connects the sections **78**. Laterally next to the connecting section **80** is a groove-shaped receiver **82** formed along the extension of the support part **28**. Further, the sections **78** each have, in the direction of extension of the support part **28**, a groove **84** and a projection **86** at the edge of the groove-shaped receiver **82** (FIG. **6**).

(138) To fix the storage devices **48** to the frame **16** and to the support parts **28** in particular, the cleaning cart **10** comprises retaining elements **70** and receiving parts **72** which latter form receiving elements **74** and cooperate with the retaining elements **70**. In the following, the receiving parts **72** are first discussed with particular reference to FIGS. **5** to **8** and **11** to **13**.

(139) The receiving part **72** is strip-shaped and extends parallel to the support part **28**, and, like the latter, it extends in the height direction **22** in particular.

(140) In a direction transverse to the height direction **22**, the receiving part **72** has an approximately L-shaped cross-section with a first side **88** and a second side **90**. Located on each side **88**, **90** is a groove **92** running in a longitudinal direction in each case.

(141) The receiving part **72** comprises on each side **88**, **90** thereof a projection **94**. The projection **94** engages behind the respective projection **86**.

(142) The receiving part **72** is, for example, made of a plastics material and can be connected to the support part **28** by hand and without tools. Here, for example, the receiving part **72** can be longitudinally slid into the support part **28** and fixed thereto.

(143) In the present case, the receiving parts **72** are dimensioned such that two receiving parts **72** arranged one above the other are held to each support part **28**.

(144) The receiving elements **74** are formed on the receiving part **72** and are each configured in the form of engagement openings **96** (FIGS. **7** and **8**). In the present case, a plurality of receiving elements **74** are arranged along the receiving part **72** in each case. These are in particular positioned equidistantly from each other.

(145) Furthermore, the receiving elements **74** are arranged such that in the two receiving parts **72**

arranged one above the other, the respective adjacent receiving elements **74** are the same distance from one another as in each of the receiving parts **72**. This results in a grid of receiving elements **74** in the height direction **22**.

(146) In each case, receiving elements **74** are arranged on either side **88** and **90**. In particular, the receiving elements **74** are in each case formed at the same location in the receiving part **72**, in the height direction **22**.

(147) As can be seen from FIGS. **3** and **5** for example, the receiving parts **72** are arranged at the spaced-apart support parts **28** such that in each case a side **88** of one receiving part **72** faces towards a side **90** of another receiving part **72**.

(148) The receiving elements **74** comprised by these are in spaced apart and opposing relation to one another.

(149) Furthermore, the receiving elements **74** are arranged on the sides of the support parts **28** that are formed by the lateral sections **78**. These two sides of the support parts are oriented transversely and in particular perpendicularly to one another.

(150) The sides **88**, **90** of the receiving parts **72** are oriented transversely and in particular perpendicularly to one another and are adjacent to one another.

(151) A respective side **88**, **90** has a contact region **98** outside the groove **92**, along the extension of the receiving parts **72**.

(152) The retaining elements **70** in the present case are formed as pullout guides **100**. Here, the retaining elements **70** are of strip-shaped configuration having a first end section **102** and a second end section **104**. The retaining element, at a respective end section **102**, **104** thereof, can be releasably fixed to two spaced-apart support parts **28**, wherein the fixing is realized indirectly via the receiving parts **72** having the receiving elements **74**.

(153) The retaining element **70** comprises a guide member **106** for a side of a container **58**, in particular for the retaining projection **62** thereof. The guide member **106** is a groove **108** in which the retaining projection **62** engages and is retained therein. The guide member **106** extends substantially the entire length of the retaining element **70**. Preferably, the container **58** can be latched with the groove **108** in at least one position.

(154) Furthermore, the retaining element **70** comprises a guide member **110** for receiving the rim **66** of the supporting element **60**. In the present case, the guide member **110** is also configured as a groove **112** which extends substantially along the entire length of the retaining element **70**. Preferably, the supporting element **60** can be latched with the groove **112** in at least one position.

(155) In particular, the grooves **108**, **112** extend parallel to one another, wherein the groove **112** for the supporting element **60** is arranged above the groove **108** for the container **58**.

(156) In intended use of the retaining elements **70**, the grooves **108**, **112** are oriented in a plane in particular transverse and specifically perpendicular to the height direction **22**, in particular horizontally.

(157) In the present case, the pullout guides **100** are simple to produce from a manufacturing standpoint in the form of moulded plastic parts which do not comprise any moving parts for guiding the storage devices **48**. In a different embodiment not depicted in the drawing, the pullout guides **100** can be telescopic, for example.

(158) The retaining element **70** has at each of its end sections **102**, **104** a retaining projection **114**. The retaining projection **114** in a connecting position can engage in the engagement opening **96** of the receiving element **74**. This gives the possibility of fixing the retaining element **70** to the respective receiving element **74** and hence indirectly to the support part **28**.

(159) As can be seen from FIG. **11** in particular, the retaining projection **114** is displaceably arranged at the retaining element **70**. The retaining element **70** forms a sleeve **116** for receiving the retaining projection **114**. Arranged in the sleeve **116** is a biasing element **118**, in the present case configured as a helical spring. The biasing element **118** applies to the retaining projection **114** a force acting in direction of the connecting position and thereby secures the latter's engagement with

the receiving element **74**.

(160) The retaining projection **114** can be moved against the action of the biasing element **118** into a release position in which the engagement with the receiving element **74** is removed. The retaining element **70** can thereby be released from the receiving element **72**. An actuating member **120** is provided for displacing the retaining projection **114** (FIG. **10**).

(161) The retaining element **70**, at the end sections **102**, **104** thereof, comprises two contact regions **122** for cooperating with the contact regions **98** at the receiving part **72**. The contact regions **122** are oriented transversely to one another and are in particular perpendicularly adjacent to one another. A first contact region **122** can make area contact and flange-like contact against the side **88**. A second contact region **122** can make area contact and flange-like contact against the side **90** and additionally against the projection **86** (FIGS. **11** to **13**).

(162) The retaining element **70** further comprises at least one orienting projection **124**. In the present case, two orienting projections **124** are arranged at one of the contact regions **122**. This is the contact region **122** which does not have the retaining projection **114** positioned therein (FIG. **10**). The orienting projections **124** can engage in the groove **92** of the receiving part **72**.

(163) The retaining elements **70** can be releasably fixed to the receiving elements **74** in a user-friendly manner, by hand and without tools. Because of the multiple receiving elements **74** in the height direction **22**, the possibility exists for the retaining elements **70**, and hence the storage devices **48**, to be fastened to the frame **16** at different elevations.

(164) Furthermore, because of the arrangement of the receiving elements **74** on both sides, the possibility exists for the retaining elements **70** to be selectively fixed on the sides **88** and **90** of the receiving parts **72**. It is thereby possible for the retaining elements to be fixed on the sides of the support part **28** in a first orientation or a second orientation of two possible orientations. These orientations are oriented transversely and in particular perpendicularly to one another and can be oriented in particular along the travel direction **34** and in the transverse direction **36**.

(165) In the present case, the retaining elements **70** are in each case used in pairs for holding the storage devices **48**. The retaining elements **70** are fixed to support parts **28** arranged opposite one another, wherein the retaining element **70** at each of the end portions **102**, **104** thereof is connected to one of the support parts **28**. A container **58** and/or a supporting element **60** can be positioned between the retaining elements **70** in the receiving space **46**.

(166) For example, in the configuration of the cleaning cart shown in FIGS. **1** to **3**, **14** and **15**, the receiving space **46** is divided such that it receives two low containers **58** and one tall container **58**. In addition, supporting elements **60** are received in each case.

(167) Absent containers **58**, for example, the supporting elements **60** can form intermediate shelves of the receiving space **46**.

(168) Furthermore, the possibility exists for a supporting element **60** and a container **58** underlying same to be held to a respective retaining element **70**, as illustrated in the drawing. In this case, the supporting elements **60** form cover elements **126** for the containers **58**, covering these substantially completely. The cover elements **126** have substantially the same cross-section as have the containers **58** at the top edges thereof.

(169) Preferably, the possibility exists for the cover elements **126** to form covers for the containers **58** for the transport thereof outside the cleaning cart **10**.

(170) The containers **58** can be displaced from a retracted position in which they are arranged in the receiving space **46** (FIG. **2**) to at least one extended position via the guide members **106**. The displacement is realized on the guide members **106** of the retaining elements **70**. For example, in FIG. **14**, the three containers **58** are each displaced to a varying extent from the receiving space **46** to the left side **42**, in the transverse direction **36**.

(171) It is advantageous for the displacement along the retaining element **70** to be possible in mutually opposite directions. For example, FIG. **15** shows the two small containers **58** as having been displaced in mutually different orientations in the transverse direction **36**, namely on the one

hand to the left side **42** and on the other to the right side **44**.

(172) Depending on what side of the cleaning cart **10** a door **50** is located on and/or whether or not the door **50** is open, versatile and user-friendly access is thereby provided to the contents of the containers **58**.

(173) It will be understood that this advantage can also be achieved when side walls or doors are absent from the cleaning cart.

(174) FIGS. **15** and **16** illustrate another configuration of the cleaning cart **10**. By comparison with the original configuration, this configuration has two of the retaining elements **70** arranged at the support parts **28** in a different orientation.

(175) Here, the retaining elements **70** are fixed to the receiving elements **74** of the respective other of the sides **88** and **90**.

(176) Depending on what orientation the retaining elements **70** are in with respect to the frame, the storage devices **48** are displaceable and extendable relative to the frame **16** in the first orientation or in the second orientation. In the present example, the two orientations are oriented transversely and in particular perpendicularly to one another.

(177) In the present example, the orientation of the retaining elements **70** for the arrangement of the lower container **58** shown in FIG. **15** is rotated 90° by comparison with the orientation of the retaining elements **70** for the arrangement as shown in FIG. **14**. This example illustrates how the lower container **58** can be moved in different directions relative to the receiving space **46** depending on the arrangement and orientation of the retaining elements **70**.

(178) Absent side walls or with doors open, in the arrangement of FIG. **15**, displacement can be realized along the travel direction **34**. In this case as well, displacement from the retracted position into two extended positions in mutually opposite directions is possible.

(179) FIG. **16** shows the cleaning cart **10** assembled in a different configuration from FIG. **15**, wherein in this case two large containers **58** are used. In this case, for example, the pair of retaining elements **70** for one of the small containers **58** can be removed from the receiving parts **72**.

(180) The drawing shows only the containers **58** as being displaced. However, the possibility exists for the supporting elements **60** to be also displaced from a retracted position to different extended positions. Here, the displacement is realized along the guide members **110**.

(181) As can be seen from FIGS. **19** to **22** in particular, the door **50**, already mentioned, comprises a door wall **128** and two fixing parts **130** arranged a distance apart from each other in the height direction **22**. The fixing parts **130** are of strip-shaped configuration and receive the door wall **128** therebetween. A tension element **132** is placed in compression and tensions the fixing parts **130** in place relative to one another.

(182) A pin-like bearing element **134** is arranged on an end side of each fixing part **130**. The bearing element **134** is received in a sleeve **136** and is biased against the frame **16** by way of a biasing element **138**, in the present case a helical spring, arranged in the sleeve **136**.

(183) Corresponding bearing elements **140** in the form of engagement openings are arranged at the frame lower part **20** and at the frame upper part **24** (FIGS. **17** and **18**).

(184) The bearing elements **134** of the two fixing parts **130** are in engagement with the bearing elements **140** in order to define a pivot axis **142** for the door **50**.

(185) The fastening mechanism for the door **50** is preferably of identical or functionally equivalent configuration to the fastening mechanism for the retaining elements **70**.

(186) The door **50** can be transferred from a closed position in which it closes off the receiving space **46** on one side, to an open position in which a user is afforded access to the receiving space **46**, and the storage devices **48** in particular.

(187) The bearing elements **140** are duplicated for each of the sides **38** to **44** and are arranged at two positions. This gives the possibility of fixing the door **50** on each of the sides **38** to **44** at two positions in different orientations on the frame **16**. This is illustratively shown in FIG. **22** in which, by comparison with FIG. **19**, the door is hinged on a different side. Here, the door **50**, on the left

side **42**, is on the one hand displaced from a location near the rear side **40** to a location near the front side **38** and, on the other hand, rotated about a horizontal axis extending in the transverse direction **36**. FIG. **22** illustrates the door **50** in an open position.

(188) The segments **54** of the side wall **52** can be held to the support parts **28** using simple structure (FIGS. **23** to **25**). To this end, strip-shaped projections **144** are arranged at the segments **54** on sides thereof facing away from one another. The projections **144** can engage in the grooves **84** (FIG. **25**). For example, the segments **54** are slid into the support parts **28** from the top before mounting the frame upper part **24**.

(189) In a direction towards the frame lower part **20** and the frame upper part **24**, a respective segment **54** has at least one projection **146** (in the present case two in number) which engages in a corresponding engagement opening **148** at the frame lower part **20** and at the frame upper part **24**. This is shown in FIGS. **23** and **24** in a segment **54** that is displaced somewhat in the height direction **22**. By virtue of the engagement of the projections **146** in the engagement openings **148**, the segments **54** are prevented from rattling, which has proven to make the cleaning cart **10** comfortable to work with.

(190) Preferably, the segments **54** themselves are also in engagement one with the other. By way of example, corresponding projections and engagement openings at a respective segment **54** are shown in FIG. **23**.

(191) The segments **54**, like the doors **50**, can be attached to each of the sides **38** to **44**.

(192) In the following, further preferred embodiments of the cleaning cart in accordance with the invention will be discussed. The advantages that have already been described in the context of the cleaning cart **10** can also be achieved with these cleaning carts. To avoid repetition, reference may be had to what has been described in the foregoing. Only the essential differences will be discussed.

(193) The same reference numerals are used for features and components that are identical or functionally equivalent.

(194) A cleaning cart **150** as shown in FIG. **26** is also provided with a door **50**. The door **50** has an extension in the height direction **22** which is less than the distance of the frame lower part **20** from the frame upper part **24**. A drawer **152** is positioned underneath the door **50**, in the receiving space **46**. In the cleaning cart **150**, a bearing element which corresponds to the bearing element **134** is arranged at one of the retaining elements **70**; this can be seen from FIG. **27** in particular. By way of example, the bearing element **140** is formed integrally on one of the end sections **102**, **104**.

(195) Furthermore, in the cleaning cart **150**, at a top side thereof, a supporting element **154** is provided which is connected to the frame upper part. The supporting element **154** is connected to the frame **16** via strut elements **156**. One of these strut elements **156** is shown in FIG. **26** of the drawing.

(196) A retaining part **158** is used for connecting the strut element **156**. The retaining part **158** engages the strut element **156** and is connected to the receiving part **72** by way of screwing, for example (FIG. **28**).

(197) In the cleaning cart **150**, the top wall **56** is illustrated in open position. A supporting element **60** is positioned such that containers **160** can be supported thereon and protrude upwardly from the receiving space **46** through the reach-through opening **26**. In this way, the containers **160** can be accessed by the user in an easy-to-handle manner.

(198) FIGS. **29** and **30** show a cleaning cart in accordance with the invention, generally designated by the reference numeral **170**, in a preferred embodiment.

(199) In the present case, the cleaning cart **170** has no doors arranged on the left side **42** thereof.

(200) Containers **58** of various sizes are received in the receiving space **46**. In particular, here, containers **58** are provided whose extensions in the height direction **22** form ratios of powers of two.

(201) The cleaning cart **170** comprises two containers **58** of low height, three containers **58** that are

each twice the aforesaid height, and one container **58** that is, again, twice the aforesaid height.

(202) Also shown in FIGS. **29** and **30** is that surplus retaining elements **70** are fixed to the receiving parts **72** and are, in this case, unused. In particular, a grid of retaining elements **70** is provided in the height direction **22** which can be assembled with containers **58** and supporting elements **60** as desired.

(203) In the configuration shown in FIG. **29**, a supporting element **60** is provided on which cleaning utensils in the form of containers **160** are positioned. The supporting element **60** here is arranged such that the containers **160** are covered at the top sides thereof when the top wall **56** is closed (FIG. **29**).

(204) FIG. **30**, on the other hand, represents a different configuration of the cleaning cart **170**. In this case, the top wall **56** has been removed, exposing the reach-through opening **26**. As in the case of the cleaning cart **150**, the supporting element **60** is positioned sufficiently high for the containers **160** to protrude through the reach-through opening **26**, thereby bringing them within easier reach of the user.

(205) In the configuration in accordance with FIG. **29**, for example, provision could be made for the top wall **56** arranged above the containers **160** to be opened, for example by pivoting as in the case of the cleaning cart **150**. In this case, the user can access the containers **160** through the reach-through opening **26**.

(206) FIG. **31** shows a cleaning cart in accordance with the invention, designated by the reference numeral **180**, in a preferred embodiment.

(207) In the cleaning cart **180**, absent a corpus **18**, the frame **16** is open. In addition, the containers **58** are omitted.

(208) In this case, provided in the receiving space **46** are only supporting elements **60**, which can have cleaning utensils positioned thereon. For example, one supporting element **60** is positioned as an intermediate shelf and, as in the case of the cleaning carts **150**, **170**, a further supporting element **60** is positioned on the top side for receipt of containers **160**.

(209) By utilizing the retaining elements **70** with guide members **106** for the containers **58**, the cleaning cart **180** can easily be configured for receipt of containers **58** when needed.

LIST OF REFERENCE CHARACTERS

(210) **10** cleaning cart **12** ground surface **14** roller **16** frame **18** corpus **20** frame lower part **22** height direction **24** frame upper part **26** reach-through opening **28** support part **30** receiver **32** handle device **34** travel direction **36** transverse direction **38** front side **40** rear side **42** left side **44** right side **46** receiving space **48** storage device **50** door **52** side wall **54** segment **56** top wall **58** container **60** supporting element **62** retaining projection **64** handle element **66** rim **68** depression **70** retaining element **72** receiving part **74** receiving element **76** arcuate section **78** section **80** connecting section **82** receiver **84** groove **86** projection **88** first side **90** second side **92** groove **94** projection **96** engagement opening **98** contact region **100** pullout guide **102** end section **104** end section **106** guide member **108** groove **110** guide member **112** groove **114** retaining projection **116** sleeve **118** biasing element **120** actuating member **122** contact region **124** orienting projection **126** cover element **128** door wall **130** fixing part **132** tension element **134** bearing element **136** sleeve **138** biasing element **140** bearing element **142** pivot axis **144** projection **146** projection **148** engagement opening **150** cleaning cart **152** drawer **154** supporting element **156** strut element **158** retaining part **160** container **170** cleaning cart **180** cleaning cart

Claims

1. Cleaning cart, comprising: a frame having a frame lower part, a frame upper part and support parts which connect the frame lower part to the frame upper part and are spaced apart from one another, rollers at the frame lower part for moving on a ground surface, storage devices for at least one of receiving therein and supporting thereon cleaning accessories, retaining elements for

- holding the storage devices, and receiving elements arranged at the support parts in a predefined position, wherein the retaining elements are selectively fixable to the receiving elements for having the storage devices assume different relative positions with respect to the support parts, wherein the storage devices are extendable relative to the frame along two directions oriented transversely to one another, depending on the orientation of the retaining elements relative to the frame.
2. Cleaning cart in accordance with claim 1, wherein a plurality of receiving elements are provided in spaced relation to one another in a height direction of the cleaning cart, and wherein the retaining elements are selectively fixable at different positions in the height direction.
 3. Cleaning cart in accordance with claim 1, wherein the receiving elements are arranged equidistantly from one another in the height direction.
 4. Cleaning cart in accordance with claim 1, wherein, at the support parts, receiving elements are provided which are arranged on two sides, oriented transversely to one another, of the support parts, and wherein the retaining elements are selectively fixable to a receiving element on a first side or to a receiving element on a second side of the two sides, with respect to different orientations of the retaining elements relative to the support part.
 5. Cleaning cart in accordance with claim 4, wherein at least one of the following applies: two orientations are provided, these being oriented transversely to one another, wherein a respective orientation defines a pullout direction for an extendable storage device; the receiving elements arranged on the two sides arranged transversely to one another are arranged at the same position in a height direction of the cleaning cart.
 6. Cleaning cart in accordance with claim 1, wherein the storage devices are or comprise containers which have or form a receiving space for cleaning utensils.
 7. Cleaning cart in accordance with claim 6, wherein at least two containers have different heights along a height direction of the cleaning cart and/or are of identical cross-section in a direction transverse to the height direction.
 8. Cleaning cart in accordance with claim 6, wherein extensions of the containers in the height direction have ratios to each other that are given by powers of two, or wherein the extensions of the containers in the height direction are integer multiples of each other.
 9. Cleaning cart in accordance with claim 1, wherein the storage devices comprise plate-shaped or substantially plate-shaped supporting elements having a respective supporting surface for cleaning utensils.
 10. Cleaning cart in accordance with claim 9, wherein the supporting elements are intermediate shelves or dividing shelves of a receiving space of the cleaning cart.
 11. Cleaning cart in accordance with claim 9, wherein at least one supporting element forms a cover element for a container.
 12. Cleaning cart in accordance with claim 11, wherein cross-sections of the supporting element and of the container in a direction transverse to a height direction of the cleaning cart are identical or substantially identical in order to completely or substantially completely cover the container.
 13. Cleaning cart in accordance with claim 9, wherein a supporting element and a container underlying same in a height direction are holdable to at least one retaining element, wherein the supporting element directly covers the container.
 14. Cleaning cart in accordance with claim 1, wherein the cleaning cart comprises a marking device for marking the storage devices.
 15. Cleaning cart in accordance with claim 1, wherein the retaining elements are or comprise pullout guides, each of which has at least one guide member for the storage devices, wherein a respective storage device is displaceable along the guide member from a retracted position to at least one extended position and vice versa.
 16. Cleaning cart in accordance with claim 15, wherein at least one of the following applies: the storage device is configured for assuming two extended positions and is transferrable along the guide member starting from the retracted position in two mutually opposite directions to a

respective extended position; the storage devices comprise retaining projections which engage in the guide members.

17. Cleaning cart in accordance with claim 16, wherein the respective storage device is latchable with the guide member when in at least one of the retracted position and the at least one extended position.

18. Cleaning cart in accordance with claim 1, wherein the receiving elements are arranged on respective sides of the support parts that face towards another support part, spaced apart from the support part.

19. Cleaning cart in accordance with claim 18, wherein at two support parts arranged a distance apart from each other receiving elements are arranged at the support parts in opposing relation to one another.

20. Cleaning cart in accordance with claim 1, wherein at least one of the following applies: a respective retaining element is holdable at end sections thereof facing away from each other to two support parts spaced apart from each other; the retaining elements in each case are arranged in pairs, wherein a retaining element arranged at the frame in a height direction of the cleaning cart is associated with a corresponding retaining element in opposing relation thereto at the frame.

21. Cleaning cart in accordance with claim 1, wherein the support parts comprise or form the receiving elements, preferably wherein the receiving elements are formed in one piece with the support parts.

22. Cleaning cart in accordance with claim 1, wherein the cleaning cart comprises receiving parts comprising the receiving elements, which receiving parts are formed separately from the support parts and are connected to the respective support parts.

23. Cleaning cart in accordance with claim 22, wherein a plurality of receiving elements are arranged at the respective receiving part.

24. Cleaning cart in accordance with claim 22, wherein the support parts are configured as profile parts and/or wherein the receiving parts are of strip-shaped configuration.

25. Cleaning cart in accordance with claim 22, wherein at least one of the following applies: the receiving parts are fixed to the respective support part by at least one of a force-locking and a form-locking connection; the receiving parts are fixable to the respective support part at least one of by hand and without tools; the receiving parts are slidable into the respective support part.

26. Cleaning cart in accordance with claim 1, wherein at least one retaining element comprises two contact regions oriented transversely to each other and arranged adjacent to each other, the retaining element being in surface contact against corresponding contact regions of the support part or of a receiving part comprising the receiving elements.

27. Cleaning cart in accordance with claim 1, wherein at least one of the following applies: the retaining elements are fixable to the receiving elements by at least one of a force-locking and form-locking connection; the retaining elements are releasably fixable to the receiving elements; the retaining elements are at least one of fixable to and releasable from the receiving elements at least one of by hand and without tools.

28. Cleaning cart in accordance with claim 1, wherein the retaining elements and the receiving elements comprise retaining projections and corresponding engagement openings for the retaining projections which are in engagement with one another when in a connecting position.

29. Cleaning cart in accordance with claim 28, wherein at least one of the following applies: the retaining projections are arranged at the retaining elements, and the receiving elements comprise or form the engagement openings; biasing elements are provided against whose action the retaining projections are bringable out of engagement with the engagement openings.

30. Cleaning cart in accordance with claim 1, wherein the retaining elements and the receiving elements comprise orienting projections and corresponding grooves for the orienting projections, these being in engagement with each other in the connected state of the retaining elements with the receiving elements.

31. Cleaning cart in accordance with claim 1, wherein the cleaning cart comprises at least one door which is transferable from a closed position to an open position and vice versa.
32. Cleaning cart in accordance with claim 31, wherein the door comprises a door wall and fixing parts between which the door wall is held or arranged, and wherein the fixing parts are pivotably held, directly or indirectly, to the frame via bearing elements.
33. Cleaning cart in accordance with claim 32, wherein a corresponding bearing element for a bearing element at the fixing part is arranged or formed at at least one of the following: at the frame lower part; at the frame upper part; at at least one retaining element.
34. Cleaning cart in accordance with claim 32, wherein the door is attachable to the cleaning cart in two different orientations and positions in which the door, rotating about parallel pivot axes in directions pointing away from each other relative to the frame, is transferable to the open position, wherein different bearing elements are provided at at least one of the frame lower part and the frame upper part, and/or wherein at least one different retaining element having a respective bearing element is provided for this purpose.
35. Cleaning cart in accordance with claim 1, wherein arranged at the frame upper part is a top wall which is transferable from a closed position to an open position and/or which is removable from the frame upper part.
36. Cleaning cart in accordance with claim 1, wherein the cleaning cart is hand-guided and comprises a handle device graspable by a user.
-